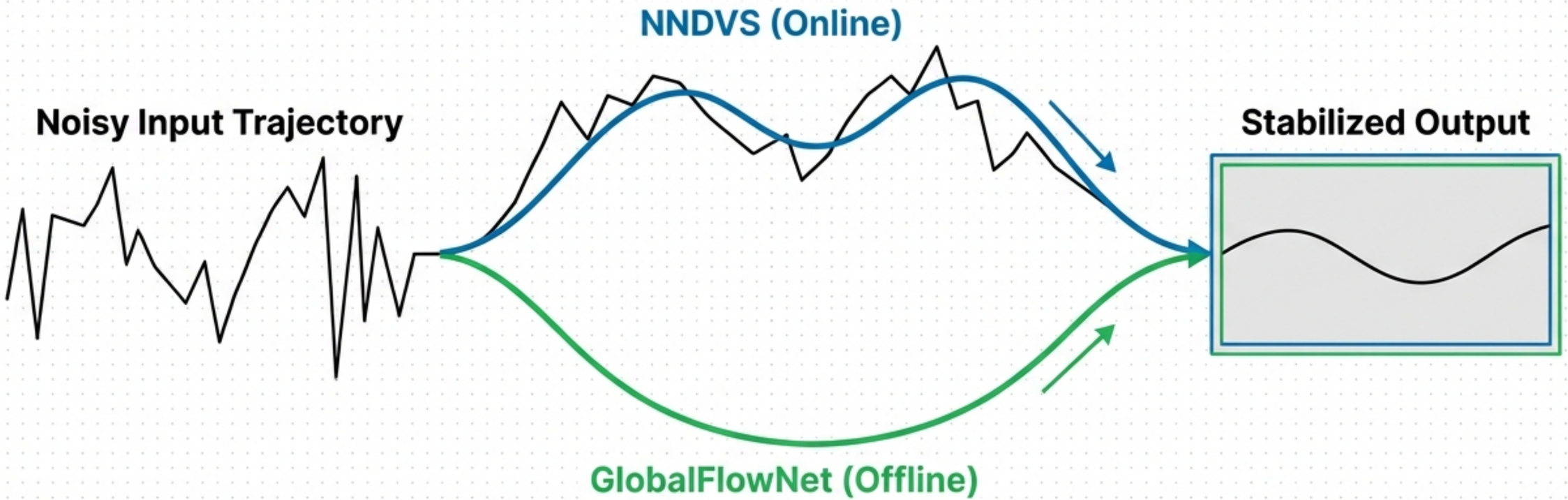
Implementing Video Stabilization

A Comparative Case Study: Online Efficiency (NNDVS) vs. Offline Precision (GlobalFlowNet)



Core Question: How do we trade latency for quality in modern mobile pipelines?

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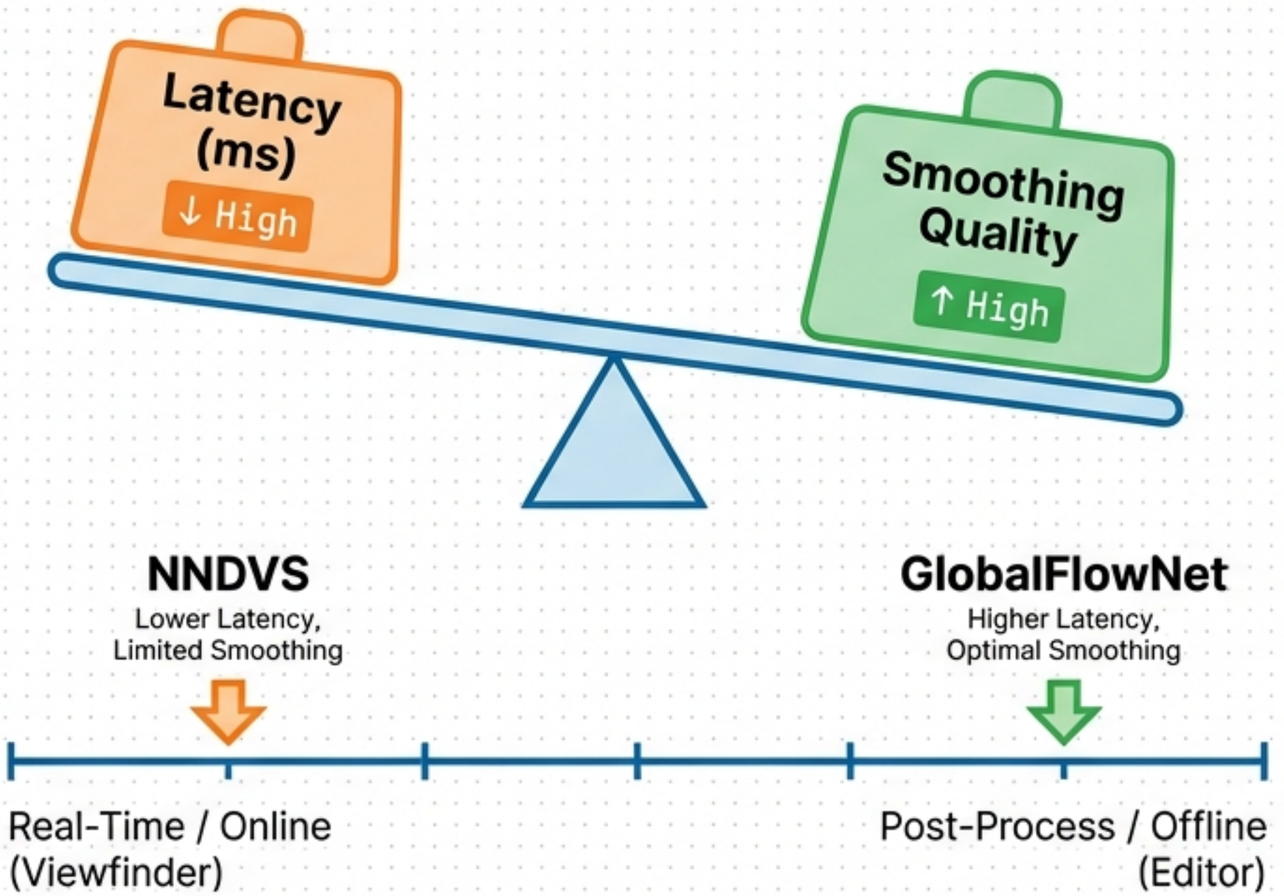
The Stability-Latency Trade-off

The Engineering Challenge

— **The Goal:** Recover a smooth camera trajectory `P(t)` from an unstable input `C(t)`.

— **The Constraint:** Mobile devices have limited thermal and compute budgets.

— **The Dilemma:** To smooth a frame, how many future frames can we see?



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The Contenders

Method A: NNDVS

Minimum Latency Deep Online Video Stabilization

- Source: `ICCV 2023`
- Type: `Online (Real-time)`
- Architecture: Hybrid (FAST/KLT + U-Net)
- Philosophy: “The Sprinter” — Fast, Sliding Window
- Target: Live Preview, Streaming

Method B: GlobalFlowNet

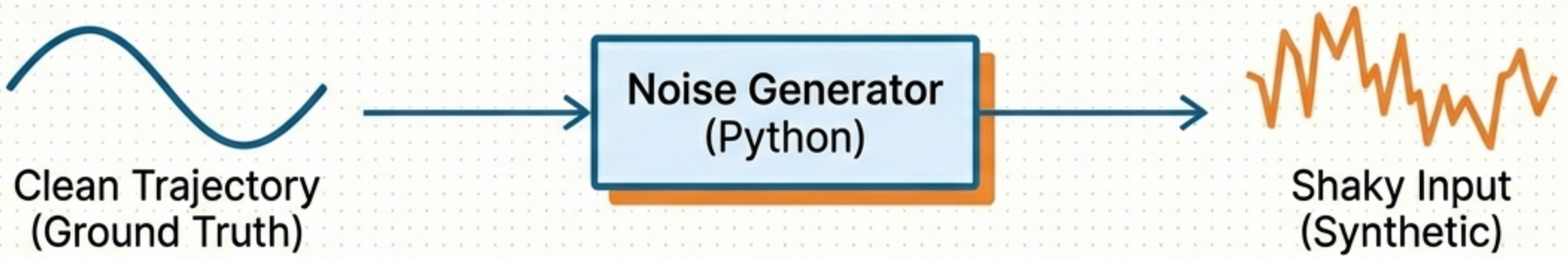
Video Stabilization using Deep Distilled Global Motion Estimates

- Source: `WACV 2023`
- Type: `Offline (Post-process)`
- Architecture: PWCNet + DCT Filtering
- Philosophy: “The Heavyweight” — Robust, Global Opt.
- Target: Gallery Editing, VFX

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Laboratory Setup & Data Generation

Testing Strategy: Synthetic Noise Injection



Clean Trajectory
(Ground Truth)

Shaky Input
(Synthetic)

Allows objective measurement against Ground Truth.

```
# 1. Clone the repository
git clone https://github.com/JeffRen1977/video_stablization.git

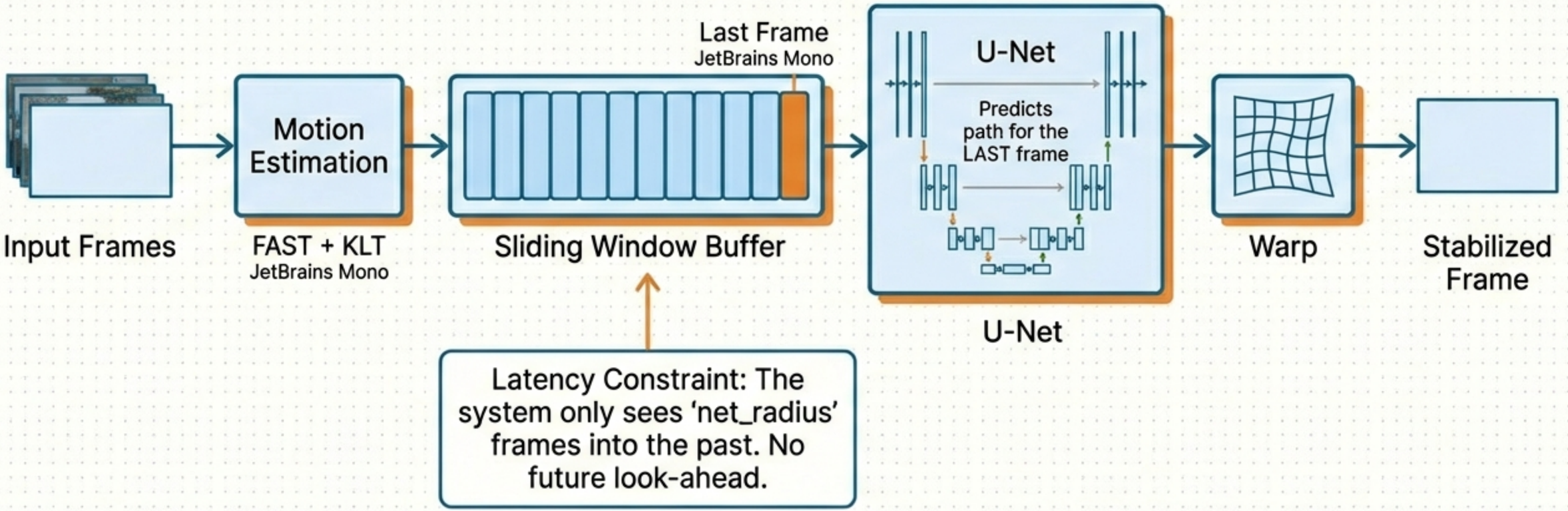
# 2. Setup environment
chmod +x setup_environment.sh

# 3. Generate synthetic shaky video for testing
python samples/stabilization/prepare_samples.py \
    --output samples/stabilization/shaky.mp4 \
    --duration 5 --fps 30
```

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Deep Dive: NNDVS Architecture

Sliding Window Processing for Real-Time Output



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NNDVS Mechanics: Hybrid Motion Estimation

Why Hybrid? Pure deep learning is too heavy for real-time motion estimation on edge devices. NNDVS uses classical CV for speed.

```
class KltTracker():

  def track_features(self, src, dst):


    # 1. FAST feature detection
    kp_src = self.detector.detect(src_gray, None)

    # 2. KLT optical flow tracking
    kp_dst, st = self.check_trace(
        src_gray, dst_gray, kp_src
    )

    # 3. Return consistent tracks
    return pts_src, pts_dst

```



Feature Detection & Tracking Visualization

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Deep Dive: GlobalFlowNet Architecture

Global Motion Estimation via PWCNet

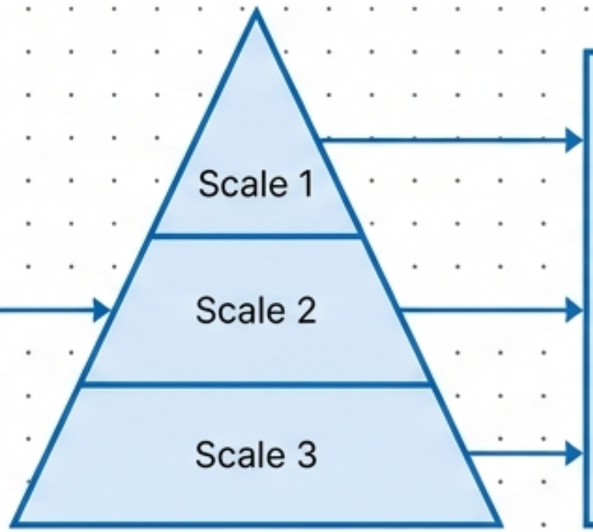
Offline Requirement: Needs access access to global context.

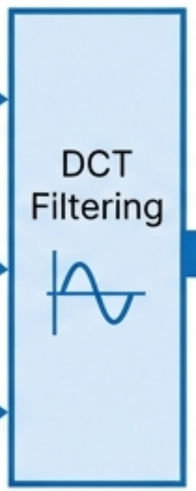


Full Video Sequence

implying offline access

PWCNet Pyramid





Global Flow Field





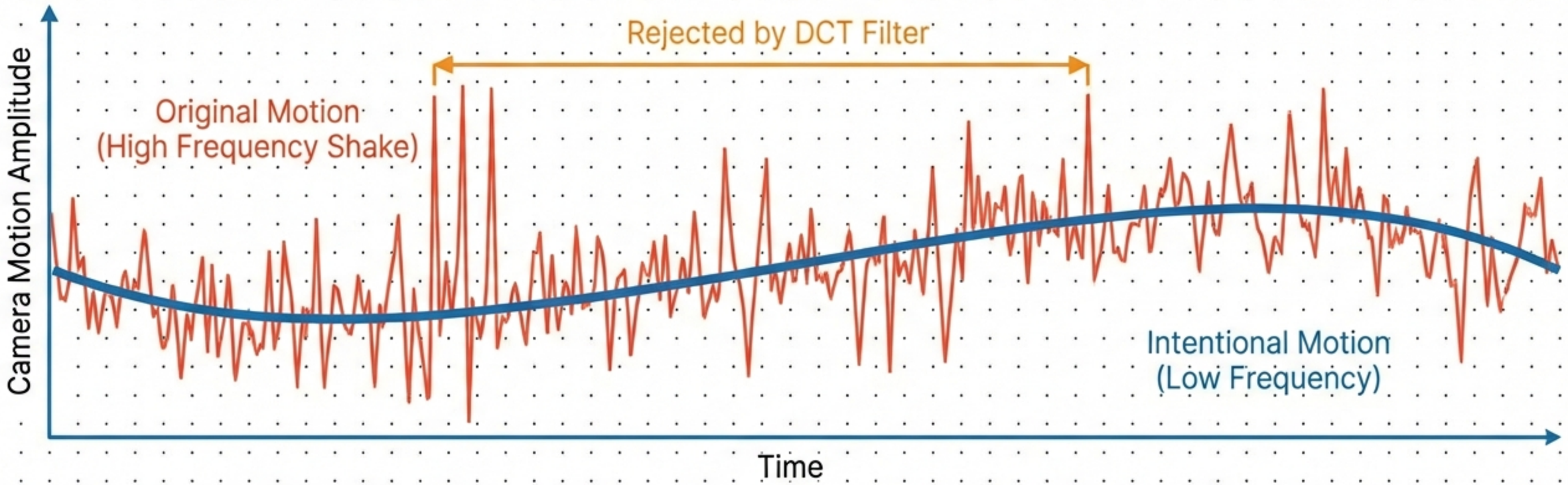
Warped Output

Result: Robust handling of complex complex scene motion.

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GlobalFlowNet: Frequency-Based Stabilization

Filtering Shake in the Frequency Domain



Basis Generation



Coefficient Extraction

Motion → Frequencies

Filtering & Reconstruction

Discard High Freq → Inverse DCT

DCT Pipeline for frequency-domain filtering of camera motion.

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Execution: Running the Algorithms

JetBrinal 1 - NNDVS / Online

```
$ bash experiments/stabilization/run_nndvs.sh \
     --ckpt thirdparty/NNDVS/pretrained_model.pth.tar \
     --out experiments/stabilization/results/nndvs_out.mp4
[INFO] Loading U-Net Model...
[INFO] Processing frame 1/150 (33ms)...
[INFO] Processing frame 2/150 (32ms)...
```

Terminal 2 - GlobalFlowNet / Offline

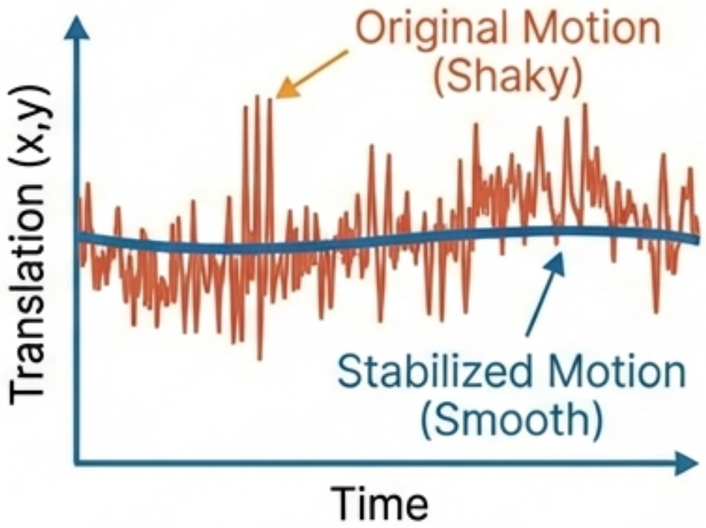
```
$ bash experiments/stabilization/run_globalflownet.sh \
    --ckpt thirdparty/GlobalFlowNet/GFlowNet.pth \
    --out experiments/stabilization/results/gflownet_out.mp4
[INFO] Loading PWCNet (37.5MB)...
[INFO] Computing Global Flow for batch size 32...
[INFO] Applying DCT Filter...
```

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Evaluation Methodology

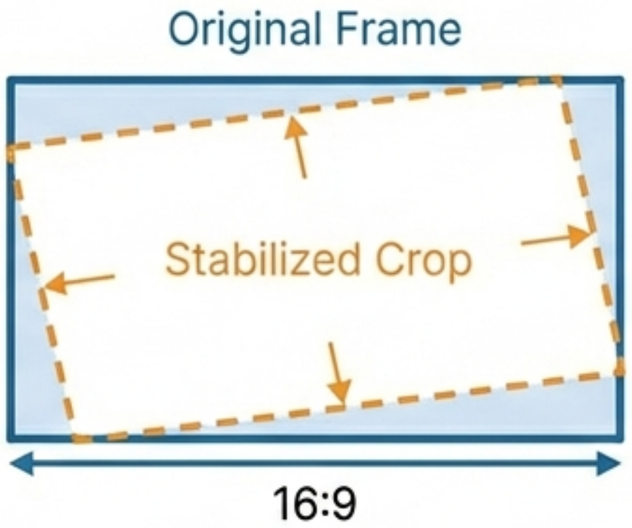
Evaluation Script: `python experiments/stabilization/eval_video.py`

Temporal Smoothness



Metric: Std Dev of Translation/Rotation.
Goal: Lower is Better.

Field of View (FOV)



Metric: Crop Ratio.
Goal: Maximize area (Less cropping).

Processing Latency



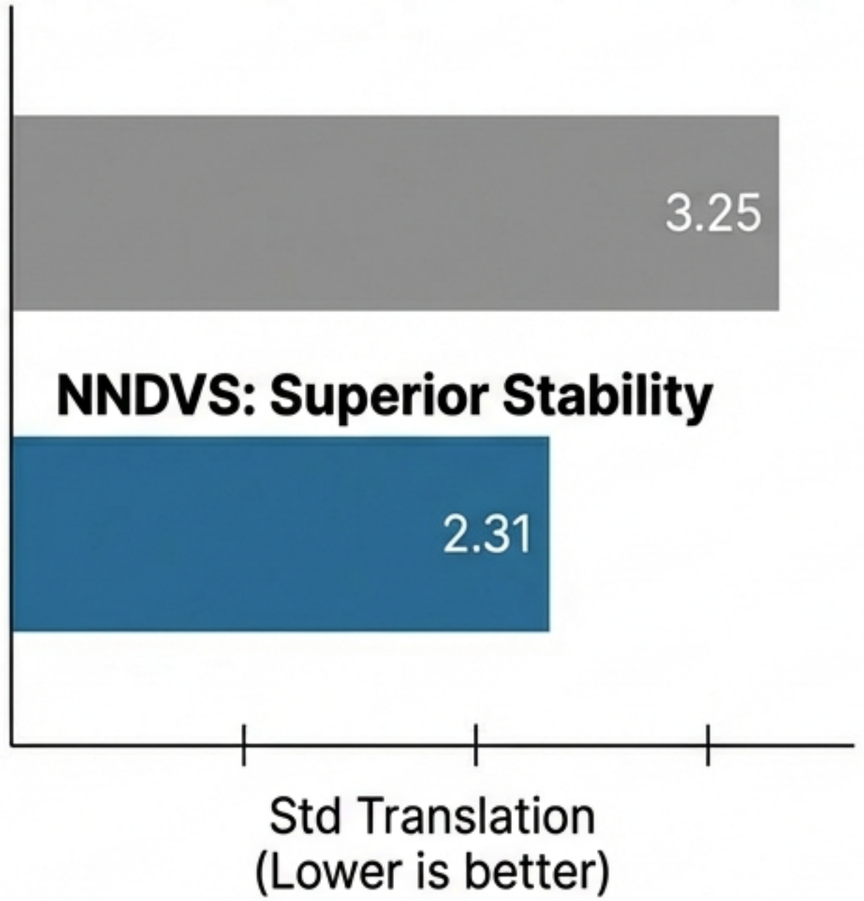
Metric: FPS / ms per frame.
Goal: >30 FPS for Real-time.

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Experimental Results: NNDVS vs. Baseline

Quantitative comparison on synthetic shaky dataset

Metric	NNDVS (Deep Learning)	OpenCV (Baseline)	Improvement
Mean Translation	5.37	7.90	32% Better
Std Translation	**2.31**	3.25	High Stability
Std Rotation	**0.72**	2.85	75% Smoother
Max Translation	10.34	15.32	Reduced Jumps



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Qualitative Analysis & Limitations

NNDVS (Online)

✅ Strengths

- Real-time capable (30 FPS on CPU).
- Robust to rotation and crowd motion.

⚠️ Limitations

- Performance dependent on window size (`net_radius`).
- Limited by accuracy of input motion estimation.

GlobalFlowNet (Offline)

✅ Strengths

- Superior global smoothing quality.
- DCT filter handles complex frequencies well.

⚠️ Limitations

- High Latency (Offline batch only).
- Heavy Model: 37.5 MB (High memory footprint).
- Requires GPU for acceptable speed.

Shared Limitation: Both methods rely on cropping (FOV loss) rather than inpainting.

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Future Improvements & Upgrade Paths

NNDVS Evolution

Current: FAST + KLT (data: `JetBrains Mono`) → **Upgrade:** Deep Optical Flow (RAFT) → **Add:** Spatial/Temporal Attention

Goal: Better feature tracking in low light.

GlobalFlowNet Evolution

Current: 37.5MB Model → **Upgrade:** Model Distillation / Compression → **Add:** Micro-batch Streaming

Goal: Deployment on mobile edge devices.

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The Verdict: Selecting the Right Architecture

Architectural choice is dictated by real-time constraints.

Application Scenario	Primary Constraint	Recommended Method
Video Conferencing	Latency (<50ms)	**NNDVS** (Online)
Electronic Viewfinder	Zero Lag	**NNDVS** (Online)
Gallery Editor	Final Quality	**GlobalFlowNet** (Offline)
VFX / Magic Eraser	Robustness	**GlobalFlowNet** (Offline)

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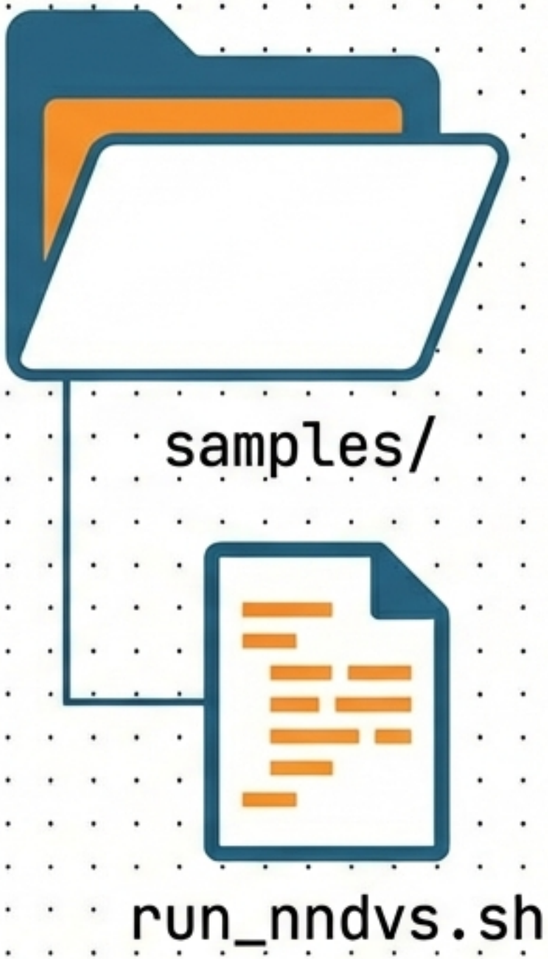
Resources & Next Steps

GitHub Repository

`https://github.com/JeffRen1977/video_stablization`

Recommended Experiments

1. **Hyperparameters**: Modify `net_radius` in NNDVS to test latency vs. smoothness.
2. **Frequency**: Adjust `cfs` (cutoff) in GlobalFlowNet to see filtering aggression.
3. **Benchmark**: Run `compare_methods.py` on your own custom footage.



References: Minimum Latency Deep Online Video Stabilization (ICCV 2023) | GlobalFlowNet (WACV 2023)

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